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### CYLINDRICAL BATTERY AND ELECTRIC DEVICE

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#### Abstract

A cylindrical battery includes a housing, an electrode assembly, and a current collector plate. The electrode assembly and the current collector plate are accommodated in the housing, and the electrode assembly includes a flattened surface, the flattened surface being provided at an end of the electrode assembly. The current collector plate includes a substrate and a bending portion interconnected. The bending portion are configured to bend towards the substrate, and the substrate is connected to the flattened surface. The substrate includes an axial hole and a plurality of through holes spaced apart from each other and running through the substrate. Along a thickness direction of the substrate, a projection area of the substrate is S1, and a sum of projection areas of the plurality of the through holes is S2, where  $4\% \leq S2/S1 \leq 16\%$ .

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/CN2023/131864, filed on Nov. 15, 2023, which claims the benefit of priority of Chinese patent application 202211441730.6, filed on Nov. 17, 2022, the contents of which are incorporated herein by reference in its entirety.

### TECHNICAL FIELD

[0002] This application relates to the field of energy storage technologies, and in particular, to a cylindrical battery and an electric device.

### BACKGROUND

[0003] In a cylindrical battery, a current collector at an end of an electrode assembly forms a flattened surface, and a current collector plate is attached to the flattened surface. A structure formed by the current collector plate and the flattened surface affects the infiltration of an electrolyte, which may lead to difficulties in replenishing electrolyte inside the electrode assembly, affecting the charge and discharge performance and the service life of the cylindrical battery.

### SUMMARY

[0004] In view of the foregoing situation, it is necessary to provide a cylindrical battery to reduce the influence of a current collector plate on the infiltration of an electrolyte.

[0005] An embodiment of this application provides a cylindrical battery including a housing, an electrode assembly, and a current collector plate, where the electrode assembly and the current collector plate are accommodated in the housing, and the electrode assembly includes a flattened surface, the flattened surface being provided at an end of the electrode assembly. The current collector plate includes a substrate and a bending portion interconnected to each other. The bending portion are configured to bend towards the substrate, and the substrate is connected to the flattened surface. The substrate includes an axial hole and a plurality of through holes spaced apart from each other and running through the substrate. Along a thickness direction of the substrate, a projection area of the substrate is  $S1$ , and a sum of projection areas of the plurality of the through holes is  $S2$ , where  $4\% \leq S2/S1 \leq 16\%$ .

[0006] In the foregoing cylindrical battery, the substrate of the current collector plate is connected to the flattened surface of the electrode assembly. The substrate is provided with the axial hole and the plurality of through holes spaced apart from each other and running through the substrate. In addition, the sum  $S2$  of the projection areas of all the through holes and the projection area  $S1$  of the substrate satisfy  $4\% \leq S2/S1 \leq 16\%$ . The plurality of the through holes on the substrate can serve as channels for an electrolyte to infiltrate the electrode assembly, reducing the influence of the current collector plate on the infiltration of the electrolyte.

[0007] In some embodiments of this application,  $5\% \leq S2/S1 \leq 10\%$ . In this case, the influence of the current collector plate on the infiltration of the electrolyte can be reduced, and the current-carrying capacity between the substrate and the flattened surface can also be improved.

[0008] In some embodiments of this application, the electrode assembly includes a positive electrode terminal and a negative electrode terminal, where along an axial direction of the

cylindrical battery, the positive electrode terminal and the negative electrode terminal are located at two ends of the electrode assembly, respectively; and the flattened surface is located at the positive electrode terminal of the electrode assembly.

[0009] In some embodiments of this application, the plurality of through holes are spaced apart along a circumferential direction of the cylindrical battery. The inventors have found that, along a radial direction of the cylindrical battery, if no through hole is provided in a region located within  $\frac{1}{5}$  to  $\frac{4}{5}$  of a radius outward from the center of the substrate, the infiltration of the electrolyte is relatively poor. Therefore, at least one through hole being provided in this region is conducive to improving the infiltration of the electrolyte on the electrode assembly.

[0010] In some embodiments of this application, the plurality of through holes are arranged in a plurality of columns, each column includes a plurality of first through holes, where the plurality of first through holes are arranged along the radial direction of the cylindrical battery to form the each column of first through holes; and the plurality of columns are spaced apart along the circumferential direction of the cylindrical battery, further improving the infiltration of the electrolyte.

[0011] In some embodiments of this application, along the radial direction of the cylindrical battery, 1 to 5 first through holes are provided in each column.

[0012] In some embodiments of this application, along the radial direction of the cylindrical battery, 2 or 3 first through holes are provided in each column.

[0013] In some embodiments of this application, a diameter of each through hole ranges from 1 mm to 3 mm.

[0014] In some embodiments of this application, the diameter of each through hole ranges from 1.3 mm to 2.5 mm.

[0015] In some embodiments of this application, the plurality of through holes include a second through hole, and when viewed along the thickness direction of the substrate, the second through hole is located between the bending portion and the axial hole.

[0016] In some embodiments of this application, at least two second through holes are provided.

[0017] In some embodiments of this application, the current collector plate has a first boundary, the first boundary being located between the substrate and the bending portion, and the bending portion are configured to bend towards the substrate along the first boundary.

[0018] In some embodiments of this application, the current collector plate further includes a first notch portion, the first notch portion connecting the substrate and the bending portion.

[0019] An embodiment of this application further provides an electric device including the cylindrical battery according to any one of the foregoing embodiments.

[0020] In the foregoing electric device, the plurality of through holes are provided on the current collector plate of the cylindrical battery. This improves the infiltration of the electrolyte on the electrode assembly, improves the charge and discharge performance and the service life of the cylindrical battery, and reduces the influence of the charge and discharge performance and the service life of the cylindrical battery on the electric device.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0021] FIG. 1 is a schematic structural diagram of a cylindrical battery according to an embodiment of this application.

[0022] FIG. 2 is a cross-sectional view of the cylindrical battery in FIG. 1 along II-II.

[0023] FIG. 3 is a schematic structural diagram of an electrode assembly according to an embodiment of this application.

[0024] FIG. 4 is a schematic structural diagram of a flattened surface formed on an electrode

assembly according to an embodiment of this application.

[0025] FIG. 5 is a schematic structural diagram of a current collector plate according to an embodiment of this application.

[0026] FIG. 6 is a cross-sectional view of the cylindrical battery in FIG. 2 along VI-VI.

[0027] FIG. 7 is a schematic structural diagram of an extended embodiment of the current collector plate in FIG. 5.

[0028] FIG. 8 is a schematic structural diagram of an extended embodiment of the current collector plate in FIG. 5.

[0029] FIG. 9 is a schematic structural diagram of an extended embodiment of the current collector plate in FIG. 5.

[0030] FIG. 10 is a schematic diagram of an electric device according to an embodiment of this application.

#### REFERENCE SIGNS OF MAIN COMPONENTS

[0031] Cylindrical battery **100** [0032] Housing **10** [0033] Electrode assembly **20** [0034] Positive electrode terminal **201** [0035] Negative electrode terminal **202** [0036] Positive electrode plate **203** [0037] Negative electrode plate **204** [0038] Separator **205** [0039] Flattened surface **206** [0040] Current collector plate **30** [0041] Substrate **301** [0042] First through hole **3011** [0043] Second through hole **3012** [0044] Axial hole **3013** [0045] Bending portion **302** [0046] First portion **3021** [0047] Second portion **3022** [0048] First boundary **303** [0049] First notch portion **304** [0050] Second boundary **305** [0051] Second notch portion **306** [0052] Output terminal **40** [0053] Conductive member **50** [0054] Electric device **200**

[0055] This application will be further described with reference to the accompanying drawings in the following specific embodiments.

#### DETAILED DESCRIPTION

[0056] The following describes the technical solutions in some embodiments of this application with reference to the accompanying drawings in these embodiments of this application. Apparently, the described embodiments are some but not all of these embodiments of this application.

[0057] It should be noted that when one component is assumed as being “connected to” another component, it may be connected to the another component directly or with a component possibly present therebetween. When one component is assumed as being “disposed on/in” another component, the component may be provided directly on/in the another component or with a component possibly present therebetween. In this application, unless otherwise specified and defined explicitly, the terms “mounting”, “connection”, “join”, and “fastening” should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or an electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, or an internal communication between two elements. Persons of ordinary skills in the art can understand specific meanings of these terms in this application as appropriate to specific situations.

[0058] In the description of some embodiments in this application, “a plurality of” means at least two unless otherwise specifically stated.

[0059] In this specification, reference to “embodiment” means that specific features, structures, or characteristics described with reference to the embodiment may be incorporated in at least one embodiment of this application. The word “embodiment” appearing in various places in the specification does not necessarily refer to the same embodiment or an independent or alternative embodiment that is exclusive of other embodiments. It is explicitly or implicitly understood by persons skilled in the art that some embodiments described herein may be combined with other embodiments.

[0060] In a cylindrical battery, a current collector at an end of an electrode assembly forms a flattened surface, and a current collector plate is attached to the flattened surface. A structure

formed by the current collector plate and the flattened surface affects the infiltration of an electrolyte, which may lead to difficulties in replenishing electrolyte inside the electrode assembly, affecting the charge and discharge performance and the service life of the cylindrical battery. [0061] An embodiment of this application provides a cylindrical battery including a housing, an electrode assembly, and a current collector plate. The electrode assembly and the current collector plate are accommodated in the housing, and the electrode assembly includes a flattened surface, the flattened surface being provided at an end of the electrode assembly. The current collector plate includes a substrate and a bending portion interconnected. the bending portion are configured to bend towards the substrate, and the substrate is connected to the flattened surface. The substrate includes an axial hole and a plurality of through holes spaced apart from each other and running through the substrate. Along a thickness direction of the substrate, a projection area of the substrate is S1, and a sum of projection areas of the plurality of the through holes is S2, where  $4\% \leq S2/S1 \leq 16\%$ .

[0062] In the foregoing cylindrical battery, the substrate of the current collector plate is connected to the flattened surface of the electrode assembly. The substrate is provided with the axial hole and the plurality of through holes spaced apart from each other and running through the substrate. In addition, the sum S2 of the projection areas of all the through holes and the projection area S1 of the substrate satisfy  $4\% \leq S2/S1 \leq 16\%$ . The plurality of the through holes on the substrate can serve as channels for an electrolyte to infiltrate the electrode assembly, reducing the influence of the current collector plate on the infiltration of the electrolyte.

[0063] The following further describes some embodiments of this application with reference to the accompanying drawings.

[0064] As shown in FIGS. 1 and 2, an embodiment of this application provides a cylindrical battery **100** including a housing **10**, an electrolyte (not shown in the figure), an electrode assembly **20**, and a current collector plate **30**. The electrolyte, the electrode assembly **20**, and the current collector plate **30** are accommodated in the housing **10**. The current collector plate **30** is connected to the electrode assembly **20**. The electrolyte can flow through a gap inside the housing **10**, and the electrolyte connects the electrode assembly **20** and the current collector plate **30**.

[0065] In an embodiment, the cylindrical battery **100** further includes an output terminal **40**. A portion of the output terminal **40** is accommodated in the housing **10** and connected to the current collector plate **30**. Another portion of the output terminal **40** is exposed on a surface of the housing **10**. In an embodiment, a portion of the output terminal **40** exceeds a surface of the housing **10**.

[0066] In an embodiment, the housing **10** includes a conductive material, and the housing **10** is connected to the electrode assembly **20**. One of the output terminal **40** and the housing **10** is a positive electrode of the cylindrical battery **100**, and the other of the output terminal **40** and the housing **10** is a negative electrode of the cylindrical battery **100**. In an embodiment, the output terminal **40** is the positive electrode of the cylindrical battery **100**, and the housing **10** is the negative electrode of the cylindrical battery **100**.

[0067] In an embodiment, the cylindrical battery **100** includes an insulating member (not shown in the figure), the insulating member connecting the output terminal **40** and the housing **10**, which is conducive to reducing the risk of a short circuit between the output terminal **40** and the housing **10**, thereby improving the safety performance of the cylindrical battery **100**.

[0068] As shown in FIGS. 2 and 3, in an embodiment, the electrode assembly **20** is a wound structure. Along an axial direction of the cylindrical battery **100**, an end of the electrode assembly **20** is a positive electrode terminal **201**, and another end of the electrode assembly **20** is a negative electrode terminal **202**. The current collector plate **30** and the output terminal **40** are located at the positive electrode terminal **201** of the electrode assembly **20**. The current collector plate **30** connects the positive electrode terminal **201** of the electrode assembly **20** and the output terminal **40**. The housing **10** is connected to the negative electrode terminal **202** of the electrode assembly **20**.

[0069] In an embodiment, the electrode assembly **20** includes a positive electrode plate **203**, a negative electrode plate **204**, and a separator **205**. The separator **205** is disposed between the positive electrode plate **203** and the negative electrode plate **204**. The positive electrode plate **203**, the separator **205**, and the negative electrode plate **204** are wound together to form a cylindrical wound structure. Along an axis direction of the wound structure, the positive electrode plate **203** on one side of the electrode assembly **20** exceeds the separator **205**, and the negative electrode plate **204** on another side of the electrode assembly **20** exceeds the separator **205**.

[0070] In an embodiment, the cylindrical battery **100** further includes a conductive member **50**, where the conductive member **50** is disposed inside the housing **10**, and the conductive member **50** connects the negative electrode terminal **202** of the electrode assembly **20** and the housing **10**. This is conducive to improving the connection stability of the negative electrode terminal **202** and the housing **10**, and reducing the risk of the negative electrode terminal **202** being disconnected from the housing **10**.

[0071] As shown in FIGS. **2**, **4**, and **5**, the electrode assembly **20** includes a flattened surface **206**, where the flattened surface **206** is formed by the positive electrode plate **203** exceeding the separator **205**. Along the axial direction of the cylindrical battery **100**, the flattened surface **206** is provided on an end surface of the positive electrode terminal **201**. The current collector plate **30** is connected to the positive electrode terminal **201** through a connection with the flattened surface **206**. The current collector plate **30** includes a substrate **301** and a bending portion **302**. The bending portion **302** are interconnected and are configured to bend towards the substrate **301**. The substrate **301** is connected to the flattened surface **206**. In an embodiment, the substrate **301** and the flattened surface **206** are connected through welding, such as ultrasonic welding or laser welding.

[0072] In an embodiment, the substrate **301** and the bending portion **302** face each other after bending; and the flattened surface **206**, the substrate **301**, the bending portion **302**, and the output terminal **40** are arranged along the axial direction of the cylindrical battery **100**. This is conducive to reducing the influence of the current collector plate **30** on space inside the housing **10** and improving the utilization of the space inside the housing **10**.

[0073] As shown in FIGS. **4**, **5**, and **6**, the substrate **301** includes an axial hole **3013** and a plurality of through holes spaced apart from each other. The axial hole **3013** and the plurality of through holes all run through the substrate **301**, so that the electrolyte is in contact connection with the flattened surface **206** through the through hole. Compared with the current collector plate **30** with no through hole on the substrate **301**, the current collector plate **30** with no through hole being attached to the flattened surface **206** and affecting the infiltration of the electrolyte on the electrode assembly **20**, the current collector plate **30** in this application allows for a direct contact connection of the electrolyte with the flattened surface **206**. This is conducive to improving the infiltration of the electrolyte on the electrode assembly **20**, and improving the charge and discharge performance and service life of the cylindrical battery **100**.

[0074] It can be understood that when the current collector plate **30** is assembled inside the cylindrical battery **100**, the axial hole **3013** is located at the axis of the cylindrical battery **100** instead of the axis of the substrate **301**. The axial hole **3013** in this application is not used for limiting a shape of the substrate **301**. Along the axial direction of the cylindrical battery **100**, a projection of the axial hole **3013** overlaps with a projection of the axis of the cylindrical battery **100**.

[0075] Along the thickness direction of the substrate **301**, that is, along the axial direction of the cylindrical battery **100**, a projection area of the substrate **301** is  $S_1$ , and a sum of projection areas of the plurality of through holes is  $S_2$ , where  $4\% \leq S_2/S_1 \leq 16\%$ . In this way, the infiltration of the electrolyte on the electrode assembly **20** can be improved. A solid portion of the substrate **301** is connected to the flattened surface **206**. This can ensure that the substrate **301** has a sufficient area for connection with the flattened surface **206**, improving the connection stability of the current collector plate **30** and the electrode assembly **20**, thereby improving the vibration resistance of the

cylindrical battery **100**. In addition, this can ensure the current-carrying capacity of the substrate **301**, reducing the influence of the through hole on the current-carrying capacity of the substrate **301**. The projection area S1 of the substrate **301** is a value, where the value is obtained by subtracting a projection area of the axial hole **3013** from an area of a region enclosed by a projection of an outer contour of the substrate **301** along the thickness direction of the substrate **301**. In an embodiment, along the thickness direction of the substrate **301**, the projection area S1 of the substrate **301** is equal to a difference between a value and the projection area of the axial hole **3013**, where the value is obtained by adding the sum S2 of the projection areas of the plurality of through holes and a projection area of the solid portion of the substrate **301**.

[0076] It should be noted that the current collector plate **30** is connected to the flattened surface **206**, and current passes through a contact region between the current collector plate **30** and the flattened surface **206**. As an area of the contact region decreases, contact resistance between the current collector plate **30** and the flattened surface **206** increases, reducing the current-carrying capacity. As the current passing through the contact region is constant, the contact resistance increases, leading to temperature rise. In this application, the sum S2 of the projection areas of all the through holes and the projection area S1 of the substrate satisfy  $4\% \leq S2/S1 \leq 16\%$ . This is conducive to reducing the influence of the current collector plate **30** on the infiltration of the electrolyte, improving the current-carrying capacity between the substrate **301** and the flattened surface **206**, reducing the temperature rise in the contact region between the current collector plate **30** and the flattened surface **206**, and reducing the influence of the temperature rise on the cylindrical battery **100**.

[0077] In an embodiment,  $5\% \leq S2/S1 \leq 10\%$ , which is conducive to further reducing the influence of the current collector plate **30** on the infiltration of the electrolyte, and improving the current-carrying capacity between the substrate **301** and the flattened surface **206**.

[0078] In an embodiment, a value of S2/S1 is any one of 4%, 5%, 6%, 7%, 7.6%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, or 16%.

[0079] In an embodiment, the current collector plate **30** has a first boundary **303**. The first boundary **303** is located between the substrate **301** and the bending portion **302**. The bending portion **302** are configured to bend towards substrate **301** along the first boundary **303**. In an embodiment, the first boundary **303** is a virtual line. In an embodiment, the first boundary **303** is a crease.

[0080] In an embodiment, the current collector plate **30** has a first groove (not shown in the figure). The first groove is located between the substrate **301** and the bending portion **302**. The bending portion **302** are configured to bend towards the substrate **301** each other along the first groove. The first groove forms the first boundary **303**. The first groove facilitates precise bending of the current collector plate **30**, reducing the influence of improper bending of the current collector plate **30** on the space inside the housing **10**.

[0081] In an embodiment, the current collector plate **30** further includes a first notch portion **304**, the first notch portion **304** connecting the substrate **301** and the bending portion **302**. The first notch portion **304** makes a connection region between the substrate **301** and the bending portion **302** form a weak region with low structural stiffness, facilitating bending. In an embodiment, when viewed along the thickness direction of the substrate **301**, the first notch portion **304** is recessed inward from an edge of the current collector plate **30**. In an embodiment, two first notch portions **304** are provided, and the two first notch portions **304** are connected to two ends of the first boundary **303**, respectively. Optionally, the first boundary **303** intersects the first notch portion **304**.

[0082] In an embodiment, when viewed along the thickness direction of the substrate **301**, the outer contour of the substrate **301** is arc-shaped, and the axial hole **3013** is located at the center of the arc. In an embodiment, when viewed along the thickness direction of the substrate **301**, the bending portion **302** is approximately rectangular, and the width of the bending portion **302** is less than the diameter of the substrate **301**. In an embodiment, when viewed along the thickness direction of the

substrate **301**, a weld joint between the substrate **301** and the flattened surface **206** is ring-shaped, which is conducive to improving the connection stability of the substrate **301** and the flattened surface **206**, and improving the vibration resistance of the cylindrical battery **100**.

[0083] In an embodiment, the plurality of through holes are spaced apart along a circumferential direction of the cylindrical battery **100**. When the substrate **301** is connected to the flattened surface **206**, the plurality of through holes spaced apart from each other in a ring shape allow the electrolyte to be in contact connection with a plurality of positions on the flattened surface **206**, which is conducive to improving infiltration uniformity of the electrolyte on the electrode assembly **20**, and enhancing the charge and discharge performance of the electrode assembly **20**.

[0084] In an embodiment, along a radial direction of the cylindrical battery **100**, at least one through hole is provided in a region located within  $\frac{1}{5}$  to  $\frac{4}{5}$  of a radius outward from the center of the cylindrical battery **100**. The inventors have found that if no through hole is provided in the region located within  $\frac{1}{5}$  to  $\frac{4}{5}$  of the radius outward from the axis of the cylindrical battery **100**, the infiltration of the electrolyte is relatively poor. Therefore, at least one through hole is provided in this region, so that the electrolyte can be in contact connection with a middle position of the radius of the flattened surface **206**, which is conducive to improving the infiltration of the electrolyte on the electrode assembly **20**.

[0085] In an embodiment, the plurality of through holes are arranged in a plurality of columns, each column includes a plurality of first through holes **3011**. Along the radial direction of the cylindrical battery **100**, the plurality of first through holes **3011** are spaced apart to form a column of first through holes **3011**. This is conducive to allowing the electrolyte to be in contact connection with the plurality of positions on the flattened surface **206**, thereby improving the infiltration uniformity of the electrolyte on the electrode assembly **20**, and improving the charge and discharge performance of the electrode assembly **20**.

[0086] Along the radial direction of the cylindrical battery **100**, at least one first through hole **3011** is provided in the region located within  $\frac{1}{5}$  to  $\frac{4}{5}$  of the radius outward from the center of the cylindrical battery **100**. In an embodiment, along the radial direction of the cylindrical battery **100**, 1 to 5 first through holes **3011** are provided in each column. In an embodiment, along the radial direction of the cylindrical battery **100**, 2 or 3 first through holes **3011** are provided in each column. In an embodiment, along the radial direction of the cylindrical battery **100**, a quantity of the first through holes **3011** in each column is any one of 1, 2, 3, 4, or 5.

[0087] In an embodiment, along the circumferential direction of the cylindrical battery **100**, a plurality of columns of first through holes **3011** are spaced apart. This is conducive to allowing the electrolyte to be in contact connection with the plurality of positions on the flattened surface **206**, thereby improving the infiltration uniformity of the electrolyte on the electrode assembly **20**, improving the charge and discharge performance of the electrode assembly **20**, and reducing impedance and internal impedance of a connection region between the substrate **301** and the flattened surface **206**. In an embodiment, the impedance includes at least one of electronic conduction impedance or electrochemical reaction impedance.

[0088] In an embodiment, a design quantity of the first through holes **3011** in each column is subject to a design specification. Assuming that a quantity of winding layers of the electrode assembly **20** is  $n$ , the thickness of a single layer of the positive electrode plate **203** after winding is  $a$ , the thickness of a single layer of the negative electrode plate **204** after winding is  $b$ , the thickness of a single layer of the separator **205** after winding is  $c$ , the diameter of the first through hole **3011** is  $d1$ , a distance between two adjacent first through holes **3011** in a same column is  $m$ , a quantity of well-infiltrated layers of an outer ring of the electrode assembly **20** is  $x$ , and the design quantity of the first through holes **3011** in each column is  $e$ ,

[00001] 
$$e = (a + b + 2 * c) * (n - 2 * x) / (d1 + m).$$

[0089] If a value of  $e$  includes a decimal, it is rounded to the nearest integer.

[0090] The quantity  $n$  of the winding layers of the electrode assembly **20** refers to a number of unit



layers forming the electrode assembly **20** by winding, where the unit layers are obtained by stacking the positive electrode plate **203**, the separator **205**, and the negative electrode plate **204**. The quantity  $x$  of the well-infiltrated layers of the outer ring of the electrode assembly **20** refers to a quantity of well-infiltrated unit layers of the outer ring in the electrode assembly **20** formed by winding the unit layers.

[0091] In an embodiment, the diameter of the first through hole **3011** is  $d_1$ , where  $1\text{ mm} \leq d_1 \leq 3\text{ mm}$ . In this way, the infiltration of the electrolyte on the electrode assembly **20** can be improved, and the connection stability of the substrate **301** and the flattened surface **206** can be improved. In an embodiment,  $1.3\text{ mm} \leq d_1 \leq 2.5\text{ mm}$ .

[0092] In an embodiment, a value of  $d_1$  is any one of 1 mm, 1.1 mm, 1.2 mm, 1.3 mm, 1.4 mm, 1.5 mm, 1.6 mm, 1.7 mm, 1.8 mm, 1.9 mm, 2 mm, 2.1 mm, 2.2 mm, 2.3 mm, 2.4 mm, 2.5 mm, 2.6 mm, 2.7 mm, 2.8 mm, 2.9 mm, or 3 mm.

[0093] In an embodiment, the plurality of through holes include a second through hole **3012**. When viewed along the thickness direction of the substrate **301**, the second through hole **3012** is located between the axial hole **3013** and the bending portion **302** and is close to the bending portion **302**. The second through hole **3012** is conducive to allowing the electrolyte to be in contact connection with the flattened surface **206** close to a region of the first boundary **303**, thereby allowing the electrolyte to be in contact connection with the plurality of positions on the flattened surface **206**, improving the infiltration uniformity of the electrolyte on the electrode assembly **20**, and improving the charge and discharge performance of the electrode assembly **20**.

[0094] In an embodiment, at least two second through holes **3012** are provided. Optionally, two second through holes **3012** are provided, and the two second through holes **3012** are spaced apart along an extension direction of the first boundary **303**.

[0095] In an embodiment, the diameter of the second through hole **3012** is  $d_2$ , where  $1\text{ mm} \leq d_2 \leq 3\text{ mm}$ . This is conducive to ensuring both the infiltration of the electrolyte on the electrode assembly **20** and the connection stability of the substrate **301** and the flattened surface **206**. In an embodiment,  $1.3\text{ mm} \leq d_2 \leq 2.5\text{ mm}$ .

[0096] In an embodiment, a value of  $d_2$  is any one of 1 mm, 1.1 mm, 1.2 mm, 1.3 mm, 1.4 mm, 1.5 mm, 1.6 mm, 1.7 mm, 1.8 mm, 1.9 mm, 2 mm, 2.1 mm, 2.2 mm, 2.3 mm, 2.4 mm, 2.5 mm, 2.6 mm, 2.7 mm, 2.8 mm, 2.9 mm, or 3 mm.

[0097] In an embodiment, along the axial direction of the cylindrical battery **100**, the projection of the axial hole **3013** overlaps with a projection of the winding center of the electrode assembly **20**. This is conducive to allowing the electrolyte to enter the winding center of the electrode assembly **20** through the substrate **301**, improving the infiltration uniformity of the electrolyte on the electrode assembly **20**, and improving the charge and discharge performance of the electrode assembly **20**.

[0098] In an embodiment, the diameter of the axial hole **3013** is  $d_3$ , where  $1\text{ mm} \leq d_3 \leq 3\text{ mm}$ . This is conducive to ensuring both the infiltration of the electrolyte on the electrode assembly **20** and the connection stability of the substrate **301** and the flattened surface **206**. In an embodiment,  $1.3\text{ mm} \leq d_3 \leq 2.5\text{ mm}$ .

[0099] In an embodiment, a value of  $d_3$  is any one of 1 mm, 1.1 mm, 1.2 mm, 1.3 mm, 1.4 mm, 1.5 mm, 1.6 mm, 1.7 mm, 1.8 mm, 1.9 mm, 2 mm, 2.1 mm, 2.2 mm, 2.3 mm, 2.4 mm, 2.5 mm, 2.6 mm, 2.7 mm, 2.8 mm, 2.9 mm, or 3 mm.

[0100] In an embodiment, the bending portion **302** includes a first portion **3021** and a second portion **3022**. The first portion **3021** is located between the substrate **301** and the second portion **3022**, and the first portion **3021** connects the substrate **301** and the second portion **3022**. The second portion **3022** are configured to bend towards the first portion **3021**.

[0101] In an embodiment, the current collector plate **30** has a second boundary **305**. The second boundary **305** is located between the first portion **3021** and the second portion **3022**. The second portion **3022** are configured to bend towards the first portion **3021** along the second boundary **305**.

In an embodiment, the second boundary **305** is a virtual line. In an embodiment, the second boundary **305** is a crease.

[0102] In an embodiment, the current collector plate **30** has a second groove (not shown in the figure). The second groove is located between the first portion **3021** and the second portion **3022**. The second portion **3022** are configured to bend towards the first portion **3021** along the second groove, and the second groove forms the second boundary **305**. The second groove facilitates precise bending of the current collector plate **30**, reducing the influence of improper bending of the current collector plate **30** on the space inside the housing **10**.

[0103] In an embodiment, the current collector plate **30** further includes a second notch portion **306**. The second notch portion **306** connects the first portion **3021** and the second portion **3022**, and the second boundary **305** intersects the second notch portion **306**. The second notch portion **306** makes a connection region between the first portion **3021** and the second portion **3022** form a weak region with low structural stiffness, facilitating bending. In an embodiment, when viewed along the thickness direction of the substrate **301**, the second notch portion **306** is recessed from an edge of the current collector plate **30** to an interior of the bending portion **302**. In an embodiment, two second notch portions **306** are provided, and the two second notch portions **306** are connected to two ends of the second boundary **305**, respectively.

[0104] In an embodiment, in the cylindrical battery **100**, the substrate **301**, the first portion **3021**, and the second portion **3022** are arranged along the axial direction of the cylindrical battery **100**, and the second portion **3022** is connected to the output terminal **40**.

[0105] As shown in FIG. 7, in an embodiment, along the radial direction of the cylindrical battery **100**, three first through holes **3011** are provided in each column. In an embodiment, three columns of first through holes **3011** are spaced apart along a circumferential direction of the substrate **301**, and a central angle between two adjacent columns of first through holes **3011** is approximately 60°. Four second through holes **3012** are provided, and the four second through holes **3012** are spaced apart along the first boundary **303**.

[0106] As shown in FIG. 8, in an embodiment, along the radial direction of the cylindrical battery **100**, three first through holes **3011** are provided in each column. Along the circumferential direction of the cylindrical battery **100**, three columns of first through holes **3011** are spaced apart, and a central angle between two adjacent columns of first through holes **3011** is approximately 90°.

[0107] As shown in FIG. 9, in an embodiment, along the radial direction of the cylindrical battery **100**, three first through holes **3011** are provided in each column. Along the circumferential direction of the cylindrical battery **100**, four columns of first through holes **3011** are spaced apart, and a central angle between two adjacent columns of first through holes **3011** is approximately 75°.

[0108] To verify the influence of the current collector plate **30** in this application on the infiltration of the electrolyte, a plurality of comparison tests were conducted. Specific information is as follows:

[0109] A plurality of cylindrical batteries **100** were selected for comparative example tests and example tests, respectively. During test of each group of comparative examples, 10 sample cylindrical batteries **100** were used. During test of each group of examples, 10 sample cylindrical batteries **100** were used. In a comparative example, a cylindrical battery **100** used a conventional current collector plate **30** with no through hole. In an Example, a cylindrical battery **100** used a design in this application.

[0110] An average value of temperature rise of 10 cylindrical batteries **100** in a specific environment in a plurality of groups of comparative examples and examples was obtained.

[0111] Temperature rises test method: In an environment of 25±0.5° C., a cylindrical battery was charged at a constant current of 2 C until voltage of the cylindrical battery reached 4.2 V, and the cylindrical battery was charged at a constant voltage of 4.2 V until the current decreased to 0.025 C and then left standing for 60 minutes. Then, the cylinder battery was discharged at a rate of 10 C until its voltage decreased to 2.5 V and then left standing for 60 minutes. During test, temperature

of the cylindrical battery **100** was collected every 10 s.

[0112] Temperature rise was calculated in a method as follows: The highest temperature among all temperatures collected was obtained. A temperature rise was obtained by subtracting an ambient temperature (25° C.) from the highest temperature.

[0113] Statistical results are shown in the following table.

TABLE-US-00001 Sum S2 of projection areas Projection of a plurality of area S1 of through holes substrate Temperature (mm.sup.2) (mm.sup.2) S2/S1 rise/° C. Comparative 0 229.04 0.00% 47.1

Example 1 Comparative 4.92 229.04 2.15% 39.8 Example 2 Example 1 9.05 229.04 4.01% 35

Example 2 12.29 229.04 5.37% 34.1 Example 3 17.38 229.04 7.59% 33.4 Example 4 22.35 229.04 9.76% 33.9 Example 5 28.71 229.04 12.54% 34.5 Example 6 35.39 229.04 15.45% 35.3

Comparative 41.48 229.04 18.11% 37.6 Example 3 Comparative 47.50 229.04 20.74% 41 Example 4

[0114] It can be seen from the foregoing table that the temperature rise of the cylindrical battery **100** in this application is significantly lower than that of a conventional cylindrical battery **100**.

[0115] To sum up, in the cylindrical battery **100** in this application, the substrate **301** of the current collector plate **30** is connected to the flattened surface **206** of the electrode assembly **20**. The substrate **301** is provided with the plurality of through holes spaced apart from each other and running through the substrate **301**. The sum S2 of the projection areas of all the through holes and the projection area S1 of the substrate **301** satisfy  $4\% \leq S2/S1 \leq 16\%$ . The plurality of through holes on the substrate **301** can serve as channels for the electrolyte to infiltrate the electrode assembly **20**, reducing the influence of the current collector plate **30** on the infiltration of the electrolyte.

[0116] As shown in FIG. **10**, an embodiment of this application provides an electric device **200** including the cylindrical battery **100** according to any one of the foregoing embodiments. The cylindrical battery **100** can supply electrical energy to the electric device **200**.

[0117] In an embodiment, the electric device **200** includes but is not limited to an unmanned aerial vehicle, an electric vehicle, an electric two-wheeler, a household appliance, and an electric tool.

[0118] In the electric device **200** in this application, the cylindrical battery **100** has relatively good charge and discharge performance, reducing the influence of deterioration in the charge and discharge performance on the electric device **200**.

[0119] In addition, persons skilled in the art can also make other changes within the spirit of this application. Certainly, these changes made according to the spirit of this application shall fall within the scope of this application.

## Claims

1. A cylindrical battery, comprising a housing, an electrode assembly, and a current collector plate; wherein the electrode assembly and the current collector plate are accommodated in the housing, and the electrode assembly comprises a flattened surface, the flattened surface being provided at an end of the electrode assembly; wherein the current collector plate comprises a substrate and a bending portion interconnected to each other, wherein the bending portion is configured to bend towards the substrate, and the substrate is connected to the flattened surface; and the substrate comprises an axial hole and a plurality of through holes spaced apart from each other and running through the substrate; and along a thickness direction of the substrate, a projection area of the substrate is S1, and a sum of projection areas of the plurality of through holes is S2, wherein  $4\% \leq S2/S1 \leq 16\%$ .

2. The cylindrical battery according to claim 1, wherein  $5\% \leq S2/S1 \leq 10\%$ .

3. The cylindrical battery according to claim 1, wherein the electrode assembly comprises a positive electrode terminal and a negative electrode terminal; wherein along an axial direction of the cylindrical battery, the positive electrode terminal and the negative electrode terminal are located at two ends of the electrode assembly, respectively; and the flattened surface is located at

the positive electrode terminal of the electrode assembly.

**4.** The cylindrical battery according to claim 1, wherein the plurality of the through holes are spaced apart along a circumferential direction of the cylindrical battery; and along a radial direction of the cylindrical battery, at least one through hole is provided in a region located within  $\frac{1}{5}$  to  $\frac{4}{5}$  of a radius outward from the axis of the cylindrical battery.

**5.** The cylindrical battery according to claim 4, wherein the plurality of through holes are arranged in a plurality of columns, each column comprises a plurality of first through holes, wherein the plurality of the first through holes are arranged along the radial direction of the cylindrical battery to form the each column of first through holes; and the plurality of columns are spaced apart along the circumferential direction of the cylindrical battery.

**6.** The cylindrical battery according to claim 1, wherein a diameter of each of the through holes ranges from 1 mm to 3 mm.

**7.** The cylindrical battery according to claim 6, wherein the diameter of each of the through holes ranges from 1.3 mm to 2.5 mm.

**8.** The cylindrical battery according to claim 1, wherein the plurality of through holes comprise a second through hole; and when viewed along the thickness direction of the substrate, the second through hole is located between the bending portion and the axial hole.

**9.** The cylindrical battery according to claim 1, wherein the current collector plate has a first boundary, the first boundary being located between the substrate and the bending portion, and the bending portion are configured to bend towards the substrate along the first boundary.

**10.** The cylindrical battery according to claim 9, wherein the current collector plate further comprises a first notch portion, the first notch portion connecting the substrate and the bending portion.

**11.** An electric device, comprising a cylindrical battery, the cylindrical battery, comprising a housing, an electrode assembly, and a current collector plate; wherein the electrode assembly and the current collector plate are accommodated in the housing, and the electrode assembly comprises a flattened surface, the flattened surface being provided at an end of the electrode assembly; wherein the current collector plate comprises a substrate and a bending portion interconnected to each other, wherein the bending portion is configured to bend towards the substrate, and the substrate is connected to the flattened surface; and the substrate comprises an axial hole and a plurality of through holes spaced apart from each other and running through the substrate; and along a thickness direction of the substrate, a projection area of the substrate is  $S_1$ , and a sum of projection areas of the plurality of through holes is  $S_2$ , wherein  $4\% \leq S_2/S_1 \leq 16\%$ .

**12.** The electric device according to claim 11, wherein  $5\% \leq S_2/S_1 \leq 10\%$ .

**13.** The electric device according to claim 11, wherein the electrode assembly comprises a positive electrode terminal and a negative electrode terminal; wherein along an axial direction of the cylindrical battery, the positive electrode terminal and the negative electrode terminal are located at two ends of the electrode assembly, respectively; and the flattened surface is located at the positive electrode terminal of the electrode assembly.

**14.** The electric device according to claim 11, wherein the plurality of the through holes are spaced apart along a circumferential direction of the cylindrical battery; and along a radial direction of the cylindrical battery, at least one through hole is provided in a region located within  $\frac{1}{5}$  to  $\frac{4}{5}$  of a radius outward from the axis of the cylindrical battery.

**15.** The electric device according to claim 14, wherein the plurality of through holes are arranged in a plurality of columns, each column comprises a plurality of first through holes, wherein the plurality of the first through holes are arranged along the radial direction of the cylindrical battery to form the each column of first through holes; and the plurality of columns are spaced apart along the circumferential direction of the cylindrical battery.

**16.** The electric device according to claim 11, wherein a diameter of each of the through holes ranges from 1 mm to 3 mm.

**17.** The electric device according to claim 16, wherein the diameter of each of the through holes ranges from 1.3 mm to 2.5 mm.

**18.** The electric device according to claim 11, wherein the plurality of through holes comprise a second through hole; and when viewed along the thickness direction of the substrate, the second through hole is located between the bending portion and the axial hole.

**19.** The electric device according to claim 11, wherein the current collector plate has a first boundary, the first boundary being located between the substrate and the bending portion, and the bending portion are configured to bend towards the substrate along the first boundary.

**20.** The electric device according to claim 19, wherein the current collector plate further comprises a first notch portion, the first notch portion connecting the substrate and the bending portion.

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